David Hyunyoo Jang

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Education

Seoul National University **Mar 2022 – Feb 2028 (Expected)**
PharmD *Seoul, South Korea*

Korea University **Mar 2019 – Aug 2021**
Coursework in Materials Science and Engineering (no degree) *Seoul, South Korea*

Research Experience

AI Protein Design Lab, Korea Advanced Institute of Science and Technology (KAIST) **Jun 2025 – Aug 2025**
Undergraduate Researcher (advisor: Gyu Rie Lee) *Daejeon, South Korea*

De Novo Binder Design for Phenytoin and Carbamazepine Sensors

- Conducted computational de novo binder-design campaigns for phenytoin and carbamazepine sensors in collaboration with Kirill Alexandrov's lab; generated and screened candidates using diffusion-based backbone sampling, LigandMPNN, and Rosetta, then delivered shortlisted designs to the collaborators.
- Consolidated the binder-design workflow into SLURM-parallelized stages with parquet-based candidate-level metric tracking; integrated AF3/Boltz-2 refolding and PLACER analysis, and implemented symmetry-aware, linker-agnostic ligand RMSD to evaluate binding poses after protein alignment. [code]
- Developed metric-guided LigandMPNN–FastRelax recycling to steer sequence sampling beyond diversification, selecting the lowest-Rosetta-ddG structure from 16 relaxed candidates per cycle; across 63 trajectories, observed decreases in mean candidate-pool Rosetta ddG ($-30.24 \rightarrow -35.13$) and LigandMPNN scores ($0.799 \rightarrow 0.674$) from cycle 1 to 32. [code]

RFdiffusion2 Atom-wise rASA Conditioning

- Extended RFdiffusion2 inference to support atom-wise relative solvent accessibility (rASA) conditioning, enabling control over which ligand atoms are buried or exposed to guide binder orientation; implemented ligand atom-name mapping, preserved global-rASA compatibility, and demonstrated the capability on a trimethyllysine (M3L) target. [PR]

Laboratory of Computational Drug Discovery, Seoul National University **Jul 2023 – Aug 2025**
Undergraduate Researcher (advisor: Juyong Lee) *Seoul, South Korea*

PREpiBind — pMHC-II binding: representation benchmark & web server

- Independently built PREpiBind, a joint-attention framework benchmarking 10 protein representations under a fixed architecture — BLOSUM62, structure-prediction embeddings (single-residue and pairwise features from AlphaFold3/Chai-1/Boltz-1), and protein language models (PLMs: ESMC/ESM3) — for pMHC class II binding on IEDB qualitative, MS, and IC50 datasets; PLMs generally led pooled performance, exceeding NetMHCIIpan-4.3, MixMHC2pred-2.0, and DeepNeo on qualitative and MS benchmarks (NetMHCIIpan-4.3 led on thresholded IC50).
- Built leakage-controlled splits and stress-tested generalization under leave-one-molecule-out and mouse (H2) held-out validation, where PLM gains narrowed, Chai-1 became competitive, and H2 rankings depended on aggregation; re-implemented DeepNeo in PyTorch (ROC-AUC 0.894 on our test set; 0.882 reported on its original test set).
- Showed that ESMC 300M matches larger PLMs (ESMC 600M, ESM3 Small/Medium) within seed variation (ROC-AUC 0.919–0.921; ESM3 Large highest at 0.927); released code (MIT license), datasets, and trained checkpoints. [preprint; code]
- Built a public web server (precomputed embeddings for 7,282 IPD-IMGT/HLA and mouse H2 chains; zero-shot) with a custom-sequence mode for non-human alleles and Colab/CLI demos; up to $\sim 69\times$ faster than command-line tools. [server]

LLMs on the Korean Pharmacist Licensing Exam

- Designed and built an end-to-end benchmark of 27 LLMs — 7 proprietary and 20 open-source (4–104B) — across six KPLE exam years (2019–2024) in Korean and English, covering question extraction (text + vision-OCR), LLM-based translation, local/API inference, and official scoring with human-percentile normalization.
- 7 models passed all years in both languages — the top reaching the top 12% of human examinees — under a logarithmic size–performance trend; developed a multi-pronged data-contamination analysis arguing against pretraining leakage as the dominant performance driver. [preprint; code]

CB1 receptor small-molecule virtual screening

- Reproduced an end-to-end structure-based virtual screen of the Enamine Hit Locator Library (460K compounds) against the CB1 receptor on an 8-GPU cluster ($\sim$12 h) — self-compiling AutoDock-GPU from source for multi-GPU execution (the prebuilt binary used 1 of 8 GPUs) — and automating batch docking, result parsing, and ranking.

Structure & Knowledge Injection into Machine Learning Lab, Seoul National University **May 2024 – Sep 2024**
Undergraduate Researcher (advisor: Jay-Yoon Lee) *Seoul, South Korea*

Semi-supervised data augmentation for RoseTTAFold2NA *collaboration with Minkyung Baek's lab*

- Implemented the training pipeline for a semi-supervised joint-loss objective that augments RF2NA with unlabeled self-distilled structures — built the mixed labeled/unlabeled data loader, per-batch loss routing, and a distributed sampler with tunable unlabeled fraction, consolidating scattered prototype scripts into one parameterized training entry point.
- Designed and ran a controlled low-label study — PDB-only vs distilled-augmentation vs joint-loss semi-supervision across 10/30/50% label-density regimes — to test whether unlabeled data improves RF2NA learning under sparse supervision.

Systems Pharmacology Lab, Seoul National University **Jun 2023 – Jul 2023**
Undergraduate Internship (advisor: Minsoo Noh) *Seoul, South Korea*

- Performed HaCaT cell culture and PGE_2 competitive ELISA to assess anti-inflammatory effects of tryptophol; conducted dose–response analysis and benchmarking against NSAIDs.

Publications & Preprints

[1] PREpiBind: Protein Representation-integrated Epitope–MHC Class II Binding Prediction
David Hyunyoo Jang, Dongwoo Kim, Untaek Hwang, Byungho Park, Yoonjoo Choi, Juyong Lee
Submitted; preprint: *bioRxiv* (2026), doi:10.64898/2026.09.15.751749

[2] Proprietary and Open-Source Large Language Models on the Korean Pharmacist Licensing Examination: A Comparative Benchmarking Study
David Hyunyoo Jang, Juyong Lee
Under review; preprint: *medRxiv* (2025), doi:10.1101/2025.04.15.25325584

Selected Research Projects

foldspace **2026 – present**
Self-initiated; evaluating pretrained folding representations as generative priors

design_cycle **2026 – present**
Self-initiated; follow-up to LigandMPNN–FastRelax recycling

Research Software & Open-Source Contributions

- **ESMFold2–AtomWorks** — integrated ESMFold2 with AtomWorks data structures and an optional Foundry training path; exact feature parity on reference fixtures and exact output parity under deterministic GPU kernels. [code]
- **RosettaCommons Foundry** — merged LigandMPNN inference improvements: per-design and ligand-interface confidence reporting, and parser overrides for AtomArray inputs. [#306; #389]
- **Upstream fixes** — AtomWorks insertion-code residue parsing, proposed and incorporated upstream in #116 [#107]; ESMFold2 multi-component CCD ligand output, the same change now present upstream [#416].
- **patinae** (Rust/WGPU molecular viewer) — fixed renderer failures on low-limit GPUs and Windows. *Merged* [#28; #29]

Presentations

[1] PREpiBind: Protein Representation-integrated Epitope–MHC Class II Binding Prediction
KIDDS 2025 Summer Workshop (Korean In silico bioDesign and Discovery Society), Poster

[2] DeepXeno: Structure-Informed Prediction of Xenotransplantation Immunogenicity *(project later developed into PREpiBind)*
2024 SNU Student-Directed Education Undergraduate Research Program, Talk (Outstanding Research Presentation)

Awards & Funding

Research Excellence Award, 2024 SNU Student-Directed Education Undergraduate Research Program **2025**
President of Seoul National University

- Awarded to two projects per fields (ten total) for excellence in research execution and outcomes.

2024 SNU Student-Directed Education Undergraduate Research Program **2024**
Faculty of Liberal Education, Seoul National University

- Selected as one of 36 projects to receive funding for undergraduate research. *(KRW 7M)*

Teaching & Leadership

SNU Alpha | Deep Learning Study Club **Spring 2022 – Winter 2024**

Executive Member, Co-Founder *College of Pharmacy, Seoul National University*

- Co-founded and helped organize a peer-led deep learning study club for pharmacy students.
- Led weekly study sessions on Natural Language Processing (Fall 2023) and PyTorch (Spring 2024), teaching deep learning concepts and applications in computational drug discovery.

Military Service **Aug 2025 – Feb 2027 (Expected)**

Team Leader, Sergeant *III Corps, Republic of Korea Army*

- Led a 10-person air-defense team, coordinating personnel and overseeing the execution of assigned duties.
- Operated and maintained air-defense radar systems in a frontline sector along the Korean DMZ.

Clinical Pharmacy Training

Seoul National University Hospital **Dec 2024 – Feb 2025**

Hospital Pharmacy Practicum *Seoul, South Korea*

- Completed rotations across 12 specialty departments, participating in medication dispensing, drug therapy case reviews, and patient counseling under pharmacist supervision.

E-Blue Pharmacy **Feb 2025 – Mar 2025**

Community Pharmacy Practicum *Suwon, South Korea*

- Reviewed prescriptions for potential medication-related problems and participated in medication therapy reviews and counseling on prescription and over-the-counter medications under pharmacist supervision.

Technical Skills

Programming & Data: Python, PyTorch, Numpy, Bash, pandas, matplotlib, Streamlit, Nginx
Modeling & Docking: RDKit, PyMOL, Chimera, AutoDock, PyRosetta, RFdiffusion (AA), LigandMPNN, AlphaFold3, ESMFold2/ESMC, AtomWorks/Foundry, TMol, Biotite
Systems & DevOps: Linux, Git, Slurm, conda/mamba, HuggingFace
Experimental Techniques: cell culture (HaCaT), buffer/reagent prep, competitive ELISA, serial dilutions, plate reader